COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE010 | **SECTION:** | CPE21S2 |
| **DATE PERFORMED:** | 08/07/26 | **DATE SUBMITTED:** | 08/07/26 |
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## ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

### 1. PROCEDURE OUTPUT

What procedures did you perform?

Create an Algorithm, Pseudocode, Flowchart, and C++ Program for the following:
1. Compute the total and average of 3 quizzes then display the total and average.

**Algorithm**
How to compute the average scores of 3 Quizzes
1. Get the score of quiz 1, quiz 2, and quiz 3
2. Add the score of quiz 1, quiz 2, and quiz 3.
3. Divide the total by 3 to get the average.

**Pseudocode**

```
START
DECLARE score1, score2, score3, total, average

DISPLAY "Enter quiz 1 score:"
INPUT score1

DISPLAY "Enter quiz 2 score:"
INPUT score2

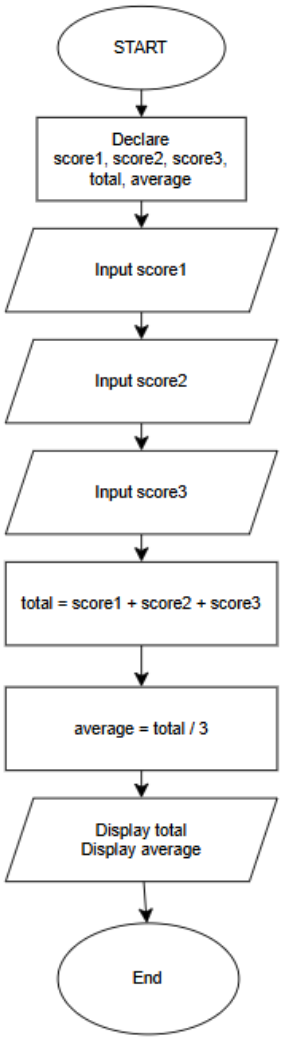
DISPLAY "Enter quiz 3 score:"
INPUT score3

COMPUTE total = score1 + score2 + score3
COMPUTE average = total / 3

DISPLAY "Total = ", total
DISPLAY "Average = ", average

END
```

**Flowchart**




**Solution**

For Task #1, I declared score1, score2, and score3 as integers because quiz scores are whole numbers. I declared average as a float since the result may contain decimal values. The program then prompts the user to enter the three quiz scores. After receiving the inputs, it computes the total by adding the three scores and calculates the average by dividing the total by 3. Finally, the program displays both the total score and the average.

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

**Algorithm**

How to make cashiering system that computes for total price of the 3 items bought, accept payment and computes change.

1. Add the prices of the 3 items
   Item 1 P200
   Item 2 P10
   Item 3 P50

2. Total purchase is P260
3. You Pay P300
4. Your change is P40

**Pseudocode**

```
START
DECLARE  item1,  item2,  item3
DECLARE  price1,  price2,  price3
INPUT item1
INPUT price1

INPUT item2
INPUT price2

INPUT item3
INPUT price3

total = price1 + price2 + price3

DISPLAY total

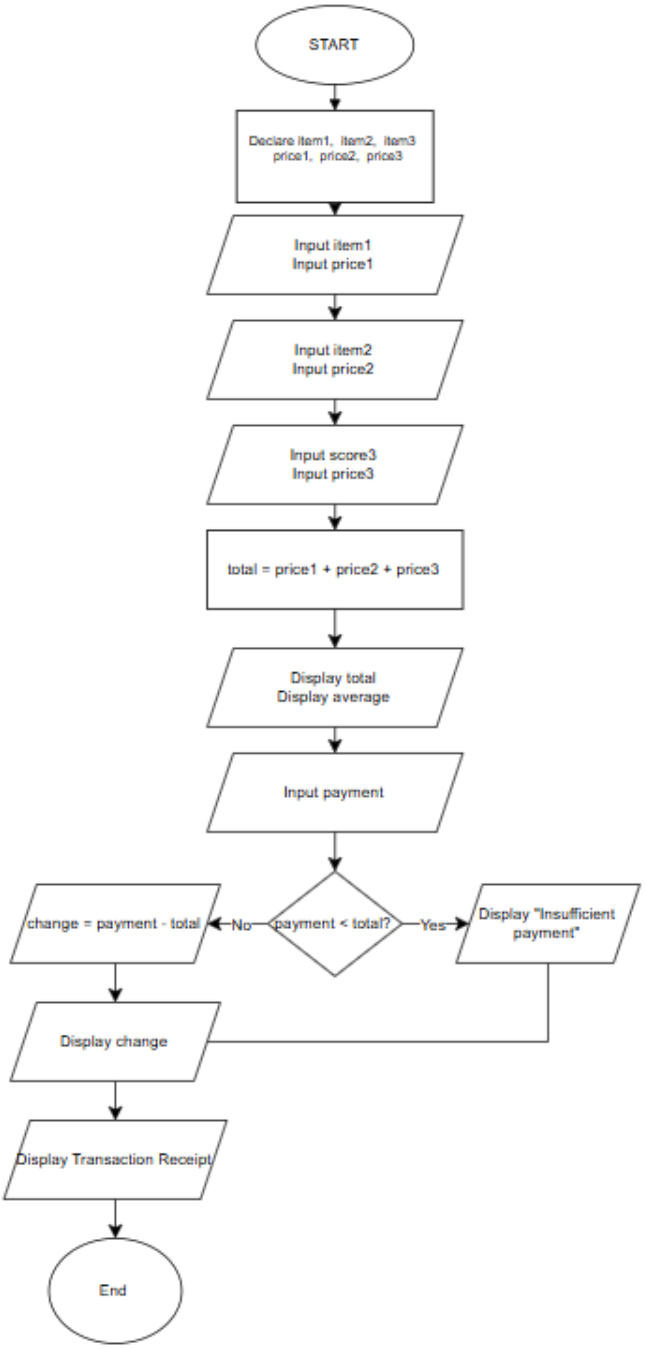
INPUT payment

IF payment < total THEN
   DISPLAY "Insufficient payment"
ELSE
   change = payment - total
   DISPLAY change
END IF

DISPLAY "Transaction Receipt"
DISPLAY item1 and price1
DISPLAY item2 and price2
DISPLAY item3 and price3
DISPLAY total
DISPLAY payment
DISPLAY change
DISPLAY "Thanks for your transaction!"

END
```

**Flowchart**




**Solution**

For Task #2, I declared item1, item2, and item3 as strings because they store the names of the items entered by the user. I declared price1, price2, price3, total, payment, and change as floats since they may contain decimal values. The program first prompts the user to enter the names and prices of the three items. It then computes the total price by adding the prices of all the items. After displaying the total price, the program asks the user to enter the payment amount. If the payment is less than the total price, it displays "Insufficient payment." Otherwise, it computes the change by subtracting the total price from the payment. Finally, the program displays a transaction

receipt containing the item names, their corresponding prices, the total amount, the payment, the change, and a thank-you message.change.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** ChatGPT, Google, QuillBot

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** Grammar correction, flowchart symbols, pseudocode guide, and code assistance

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** I used the following tools (ChatGPT, Google, Quillbot) to make sure my grammar is correct. I also reviewed the flowchart symbols and how to make pseudocode since I somehow forgot, and I made ChatGPT explain how I can fix my code since I encountered some errors while doing the tasks.

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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